CHITTI BHASKARA RAO

Vlsi Verification Engineer (Fresher) | IoT Application Development | Microcontrollers | C | Hardware Interfacing
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PROFILE SUMMARY

Final year B.Tech ECE student at Lingaya's Institute of Management and Technology (CGPA: 7.2) with 6 months of hands-on internship in VLSI Design & Semiconductor at Blackbucks IIDT and 2 months at Skill Dzire. Certified in IoT (NPTEL) with strong understanding of microcontrollers, hardware-software interfacing, and sensor-based applications. Proficient in C programming fundamentals and embedded system design concepts.

TECHNICAL SKILLS

Embedded Systems: Microcontroller Programming, Hardware-Software Interfacing, Real-Time Systems, Sensor Integration
IoT Development: IoT Application Design, Communication Modules, Smart Device Integration, M2M Communication
Programming: C Programming, Python (Basics), Scripting, MATLAB (Image Processing)
Electronics Fundamentals: Digital Electronics, Logic Design, CMOS Concepts, Circuit Analysis
Tools & Platforms: Embedded Development Tools, NASCOM Data Acquisition, IoT Platforms, MATLAB
VLSI (Additional): CMOS Digital Design, Verilog Basics, GVIM Basics
Soft Skills: Communication, Teamwork, Analytical Thinking, Fast Learner
Languages: English, Telugu, Hindi

ACADEMIC PROJECT

Vehicle Number Plate and Speed Detection Using Image Processing in MATLAB | *Main Project | Duration: 6 Months*

- Designed and implemented a MATLAB-based image processing system to automatically detect vehicle number plates and estimate vehicle speed from video/image frames captured by a roadside camera.
- Applied image pre-processing techniques such as grayscale conversion, noise removal, and contrast enhancement to prepare frames for accurate plate localization.
- Used edge detection and morphological operations (dilation, erosion) to isolate the number plate region from the vehicle image.
- Implemented character segmentation and Optical Character Recognition (OCR) to extract alphanumeric characters from the detected number plate.
- Developed a frame-difference and distance-time based algorithm to calculate vehicle speed by tracking the vehicle's position across consecutive video frames.
- Compared computed vehicle speed against a defined speed limit and flagged over-speeding vehicles, storing the extracted plate number and speed data for reporting.
- Tested the system on multiple sample video datasets under varying lighting and background conditions to validate detection accuracy.
- Gained practical exposure to MATLAB Image Processing Toolbox, digital image processing fundamentals, and real-world traffic surveillance applications.

INTERNSHIP EXPERIENCE

Embedded System and IoT Application – Blackbucks IIDT | *Duration: 2 Months | Domain: IoT Applications*

- Worked with embedded systems concepts such as microcontroller programming, hardware interfacing, and peripheral communication.
- Developed practical knowledge of IoT applications by integrating sensors and connected devices for data collection and remote monitoring.

VLSI Design & Semiconductor Intern – Blackbucks IIDT | *Duration: 6 Months | Domain: VLSI Design & Semiconductor*

- Completed an intensive 6-month VLSI Design and Semiconductor program covering end-to-end chip design concepts.
- Gained hands-on experience in CMOS digital design, RTL design flow, logic synthesis, and timing analysis.
- Studied semiconductor device physics including MOSFETs, BJTs, CMOS fabrication, and process nodes.

VLSI Design Intern – Skill Dzire | *Duration: 2 Months | Domain: VLSI Design*

- Studied CMOS digital logic design, combinational and sequential circuits, and RTL design basics.
- Enhanced understanding of hardware design principles that complement embedded systems work.

EDUCATION

B.Tech – Electronics & Communication Engineering (ECE) | *Lingaya's Institute of Management and Technology | 2022 – 2026*
CGPA: 7.2

12th Standard – BIEAP | *Year: 2022 | Medium: English*
Grade: 87.0%

10th Standard – Andhra Pradesh Board | *Year: 2020 | Medium: English*
Grade: 100%

CERTIFICATIONS

- IoT – NPTEL Certificate
- CMOS Digital VLSI Design – NPTEL Certificate
- Acquiring Data in NASCOM
- Data Analytics Job Simulation – Deloitte
- Cyber Job Simulation – Deloitte

EXTRA-CURRICULAR ACTIVITIES

- Playing Cricket, Kabaddi – Demonstrates strong teamwork, discipline, and leadership under pressure.